

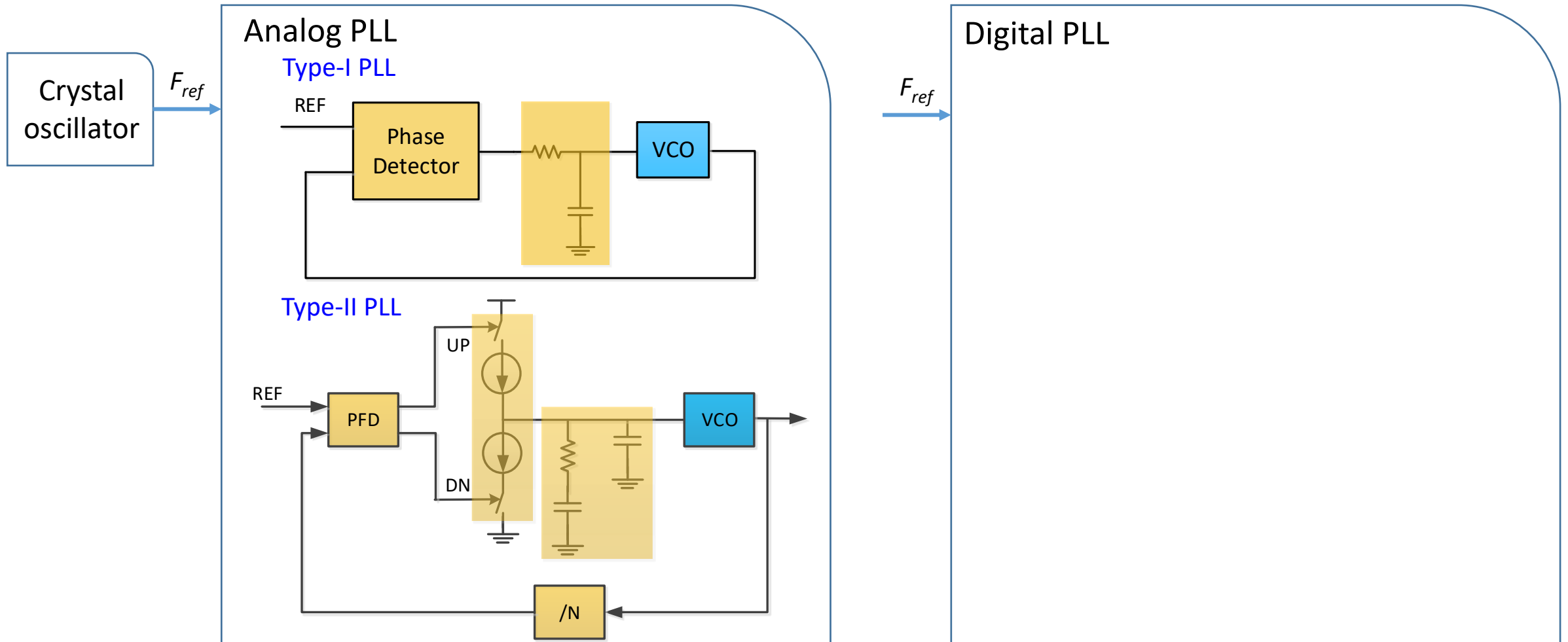
ECE264D: CMOS Analog Integrated Circuits and Systems IV

Lecture 7: Coupled Oscillators, Ring Oscillators and Millimeter Wave VCOs

Yiwu Tang



Frequency Synthesizer Overview

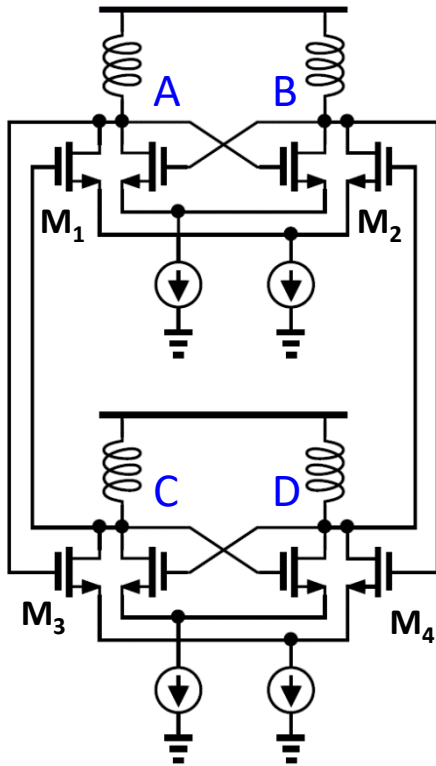


Outline

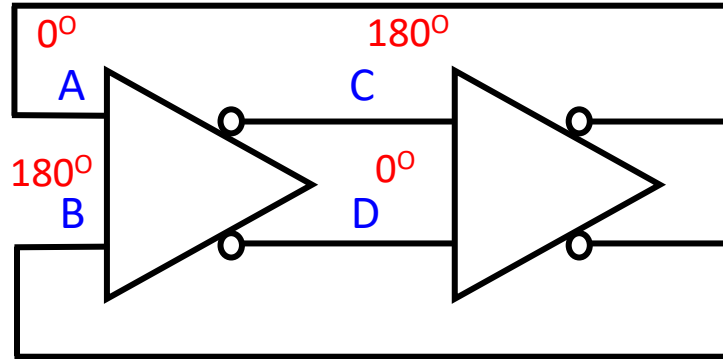
- Coupled oscillator
 - In phase multi-core oscillator
 - Quadrature and multi-phase oscillator
- Ring oscillator
 - Oscillation startup condition
 - Oscillation frequency tuning
- Millimeter wave oscillator
 - Design challenges at millimeter wave frequency
 - Push-Push and multi-push oscillator



Coupled Oscillator



In phase coupling



- Feedback model

$$X = (\alpha_1 Y - X)H(s) \quad (1)$$

$$Y = (\alpha_2 X - Y)H(s) \quad (2)$$

$$\text{From (1), } XY = (\alpha_1 Y^2 - XY)H(s) \quad (3)$$

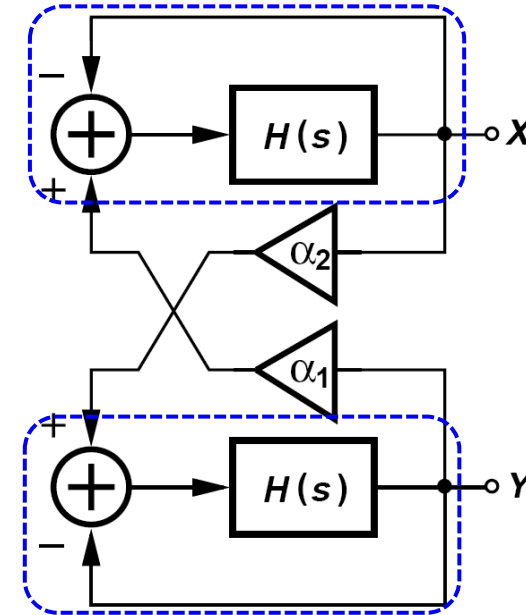
$$\text{From (2), } XY = (\alpha_2 X^2 - XY)H(s) \quad (4)$$

From (3) and (4)

$$\alpha_2 X^2 = \alpha_1 Y^2 \quad (5)$$

- For in phase coupling, $\alpha_1 = \alpha_2$,

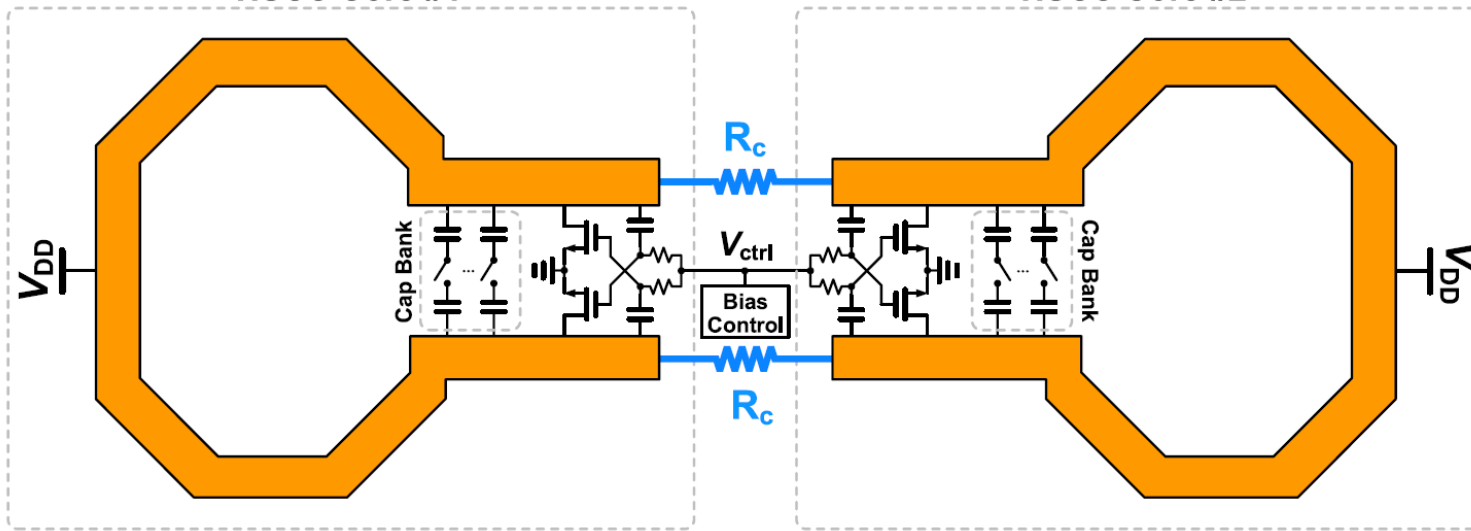
- $X = \pm Y$, X and Y are in phase or 180° out of phase.



Ref [1]



Coupled Oscillator



Ref [2]

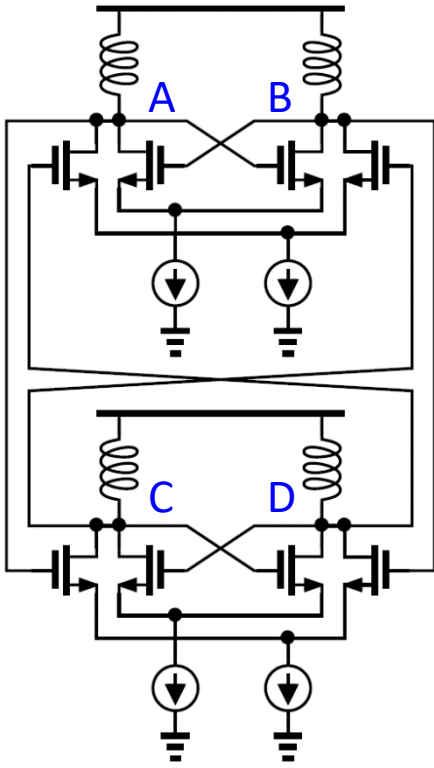
$$L_N(\Delta\omega) = L_1(\Delta\omega) - 10 \log(N)$$

$N=2$ in dual core VCO case

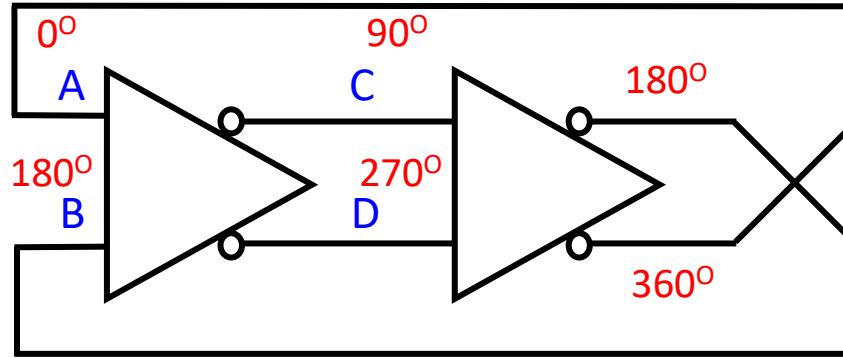
- Direct in-phase coupling
- Noise power (uncorrelated) $\uparrow 3\text{dB}$, signal power (correlated) $\uparrow 6\text{dB}$, phase noise improves 3dB
- Total current consumption doubles
- FOM remains the same
- To achieve ultra-low phase noise performance



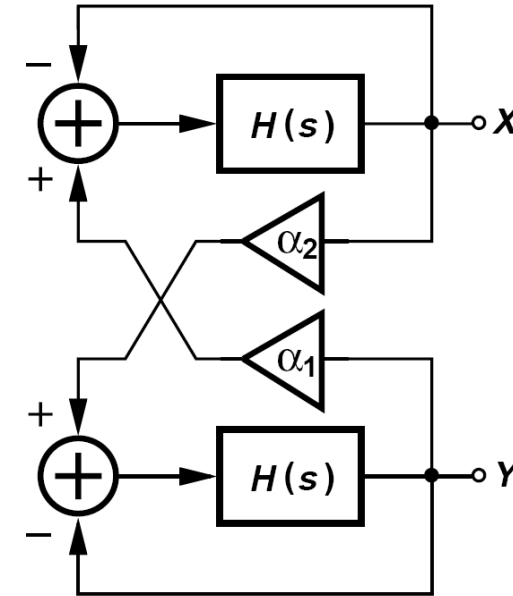
Coupled Oscillator



Anti-phase coupling



- Feedback model



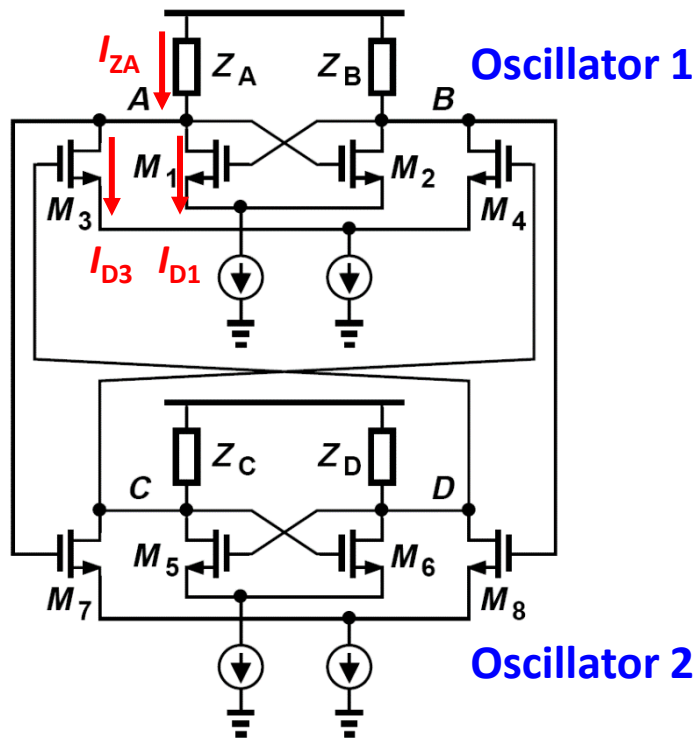
$$\begin{aligned} X &= (\alpha_1 Y - X)H(s) \\ Y &= (\alpha_2 Y - Y)H(s) \\ \alpha_2 X^2 &= \alpha_1 Y^2 \end{aligned}$$

- For anti-phase coupling, $\alpha_1 = -\alpha_2$,
 - $X = \pm jY$, X and Y are 90° out of phase $\Rightarrow$ Quadrature oscillator

Ref [1]

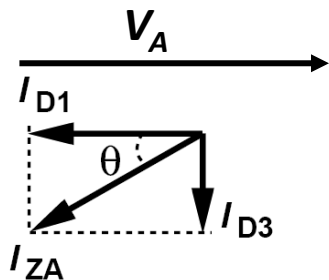


Coupled Oscillator



Oscillator 1

Oscillator 2



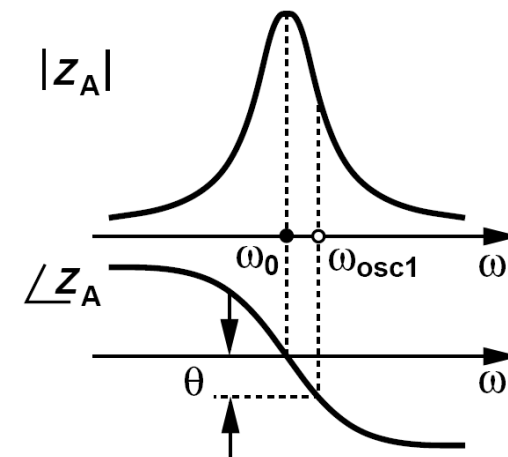
Phasor diagram

From phasor diagram

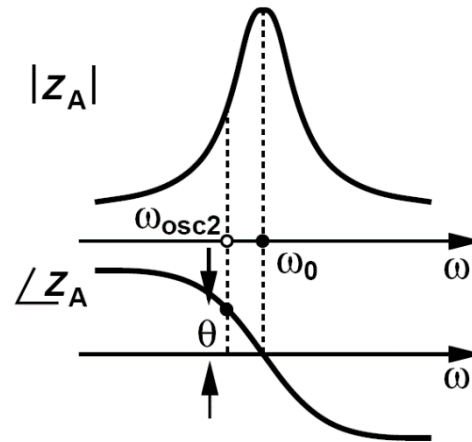
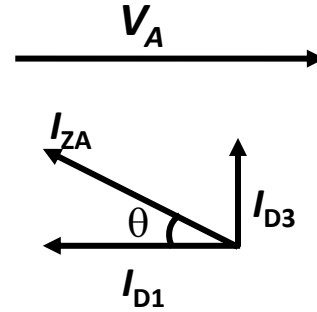
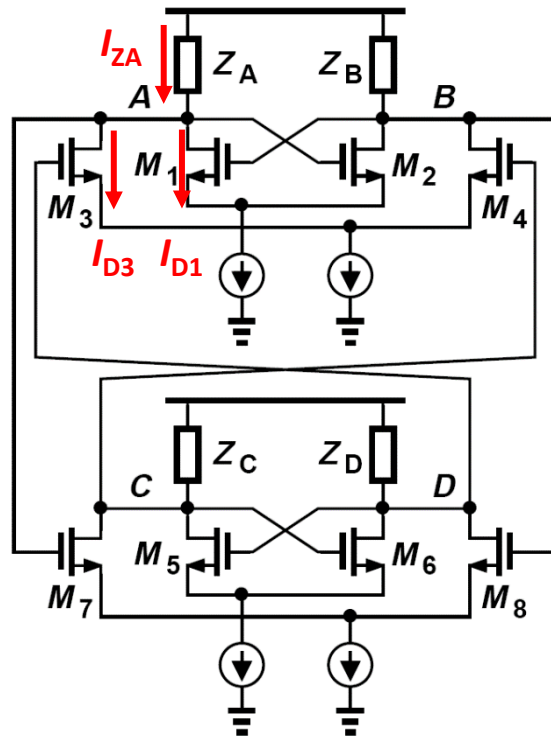
- $\vec{I}_{ZA} = \vec{I}_{D1} + \vec{I}_{D3}$
- $\vec{V}_A = -\vec{Z}_A \cdot \vec{I}_{ZA}$

Since I_{ZA} and V_A has a phase difference of θ , Z_A needs to be complex with a phase of θ .

- Oscillation frequency is different from tank resonant frequency ω_0 .
- $Z_A = (sL) // R_P // \frac{1}{sC}$
- $Z_A = \frac{j\omega_{osc}L}{1 - \omega_{osc}^2 LC + j\frac{\omega_{osc}L}{R_P}}$
- $\angle Z_A = \frac{\pi}{2} - \tan^{-1} \left(\frac{\frac{\omega_{osc}L}{R_P}}{1 - \omega_{osc}^2 LC} \right) = \tan^{-1} \left(\frac{1 - \omega_{osc}^2 LC}{\frac{\omega_{osc}L}{R_P}} \right)$
- $\omega_0 = \frac{1}{\sqrt{LC}}, \omega_{osc} = \omega_0 + \Delta\omega$
- $\angle Z_A = \tan^{-1} \left(\frac{1 - \left(\frac{\omega_{osc}}{\omega_0}\right)^2}{\frac{\omega_{osc}L}{R_P}} \right) \approx \tan^{-1} \left(\frac{\frac{-2\Delta\omega}{\omega_0}}{\frac{\omega_{osc}L}{R_P}} \right) \approx -\frac{2R_P\Delta\omega}{\omega_0^2 L} = -2\frac{Q}{\omega_0}\Delta\omega$
- Let $\angle Z_A = \theta = -\tan^{-1} \frac{I_{D3}}{I_{D1}} \rightarrow \Delta\omega = \frac{\omega_0}{2Q} \tan^{-1} \frac{I_{D3}}{I_{D1}} > 0$



Coupled Oscillator

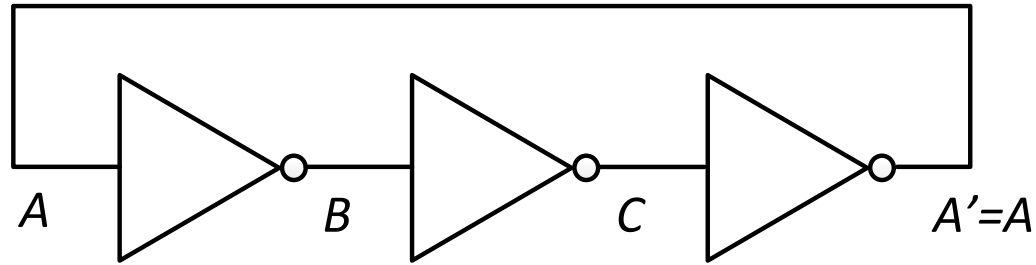


$$\Delta\omega = \frac{\omega_0}{2Q} \tan^{-1} \frac{I_{D3}}{I_{D1}} < 0$$

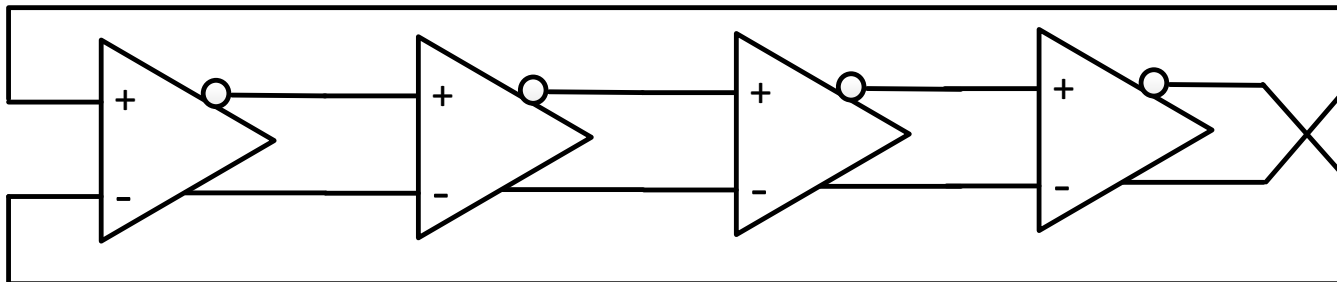
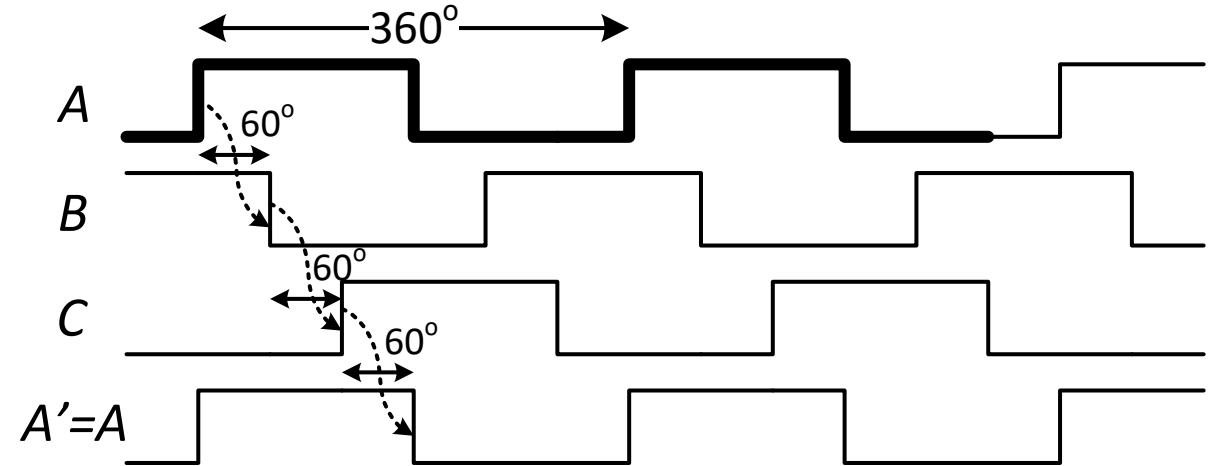
- If I_{D1} is leading I_{D3} in phase, Z_A phase needs to have the opposite sign of θ for oscillation, resulting in another oscillation frequency ω_{osc2} .
- Both oscillation modes are possible in simulation and on silicon. Circuit technique needed to enhance one mode and stabilize the oscillation [Ref 10].



Ring Oscillators



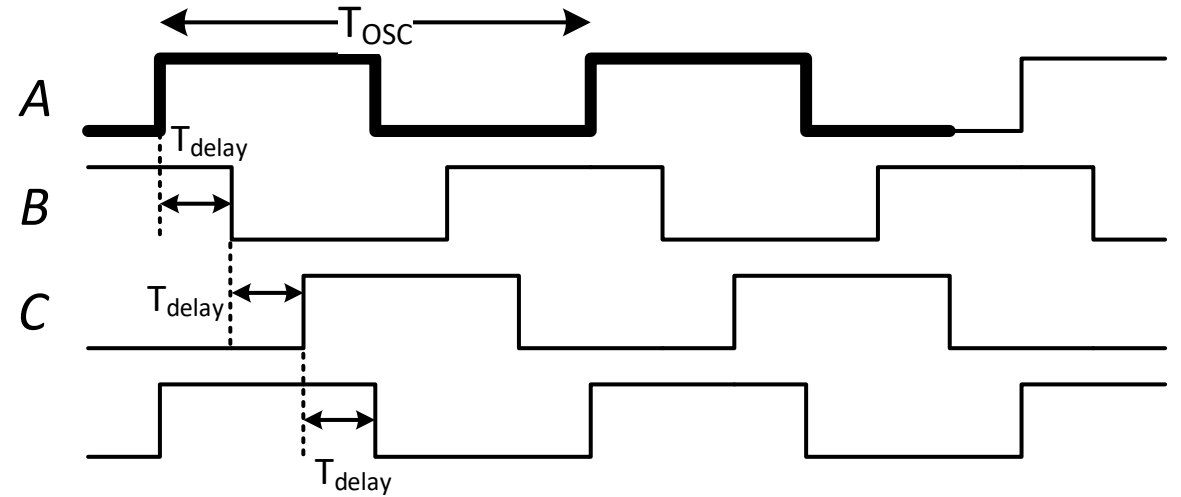
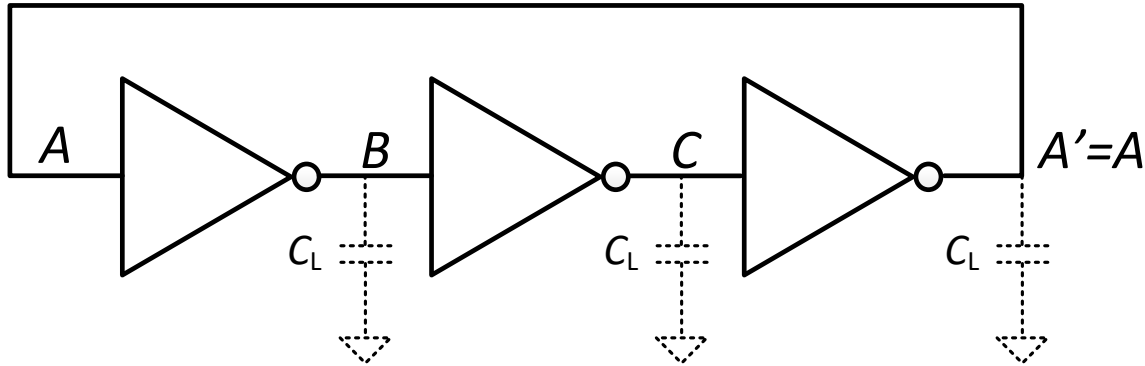
- Single-ended ring oscillator
- Each stage generates 60° delay
- Together with 180° inversion, total of 360° phase shift
⇒ positive feedback for oscillation
- Odd number of stages



- Differential ring oscillator
- Even or odd number of stages
- Proper connection to ensure oscillation



Ring Oscillators



- Total n stage inverter delay = half oscillation period

$$\frac{1}{2}T_{osc} = n \cdot T_{delay}, f_{osc} = \frac{1}{2n \cdot T_{delay}}$$

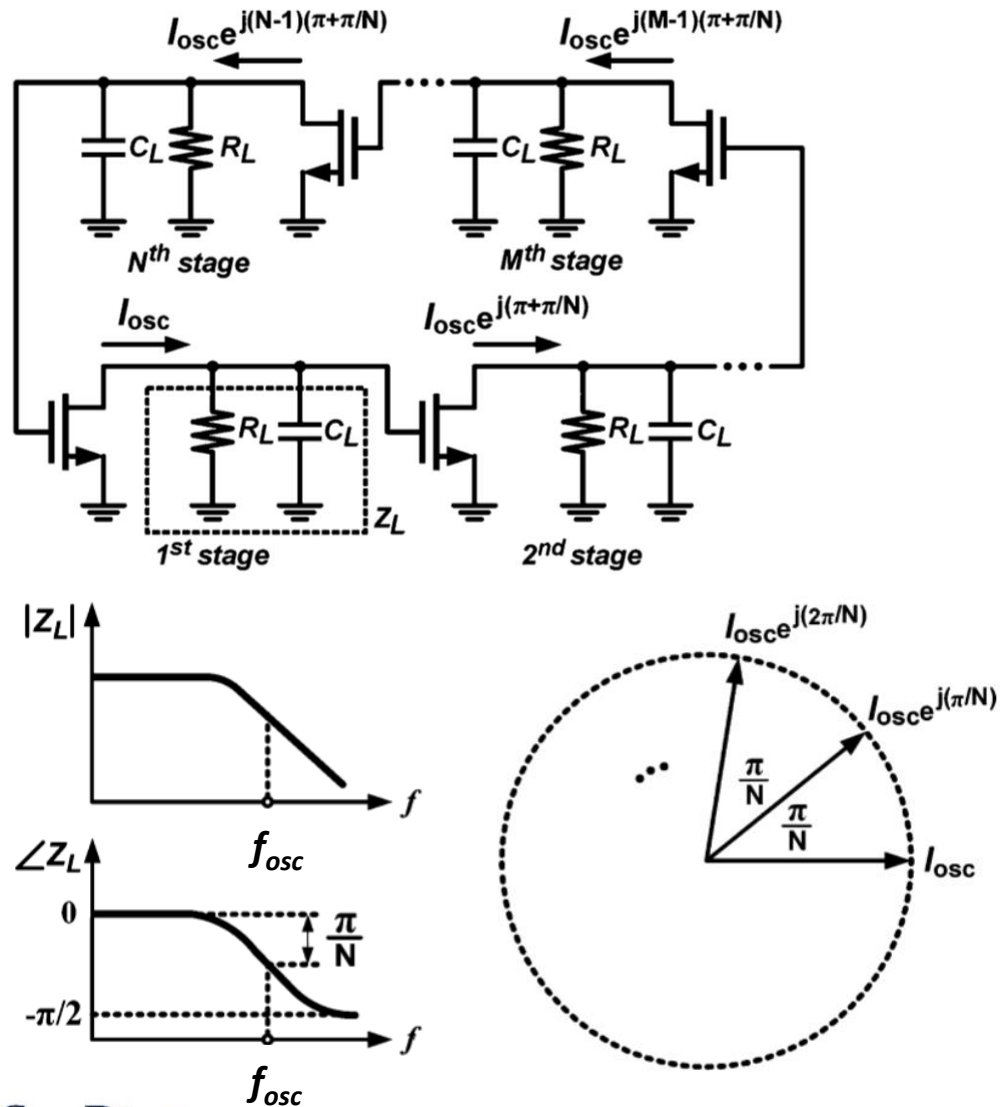
- Power consumption estimate

$$T_{delay} = \frac{C_L \left(\frac{1}{2}V_{DD} \right)}{I_{avg}}, I_{avg} = \frac{C_L V_{DD}}{2T_{delay}} = n C_L V_{DD} f_{osc}$$

$$P = n I_{avg} V_{DD} = n^2 C_L V_{DD}^2 f_{osc}$$



Ring Oscillator Startup Condition



- Barkhausen criteria for oscillation

$$|H(s = j\omega_{osc})| = 1, \angle H(s = j\omega_{osc}) = 180^\circ$$

- Each stage is a single pole system

- Gain of each stage: $H_1(s) = -g_m \left(R_L // \frac{1}{sC_L} \right) = -\frac{A_0}{1 + \frac{s}{\omega_{RC}}}$

- Where $A_0 = g_m R_L$ is the low frequency gain,

- $\omega_{RC} = \frac{1}{R_L C_L}$ is the 3-dB bandwidth of each stage

- Overall loop gain $H(s) = -\left(\frac{A_0}{1 + \frac{s}{\omega_{RC}}} \right)^N = -\left(\frac{A_0}{1 + j\frac{\omega}{\omega_{RC}}} \right)^N$

- $\angle H(s = j\omega_{osc}) = 180^\circ \rightarrow$

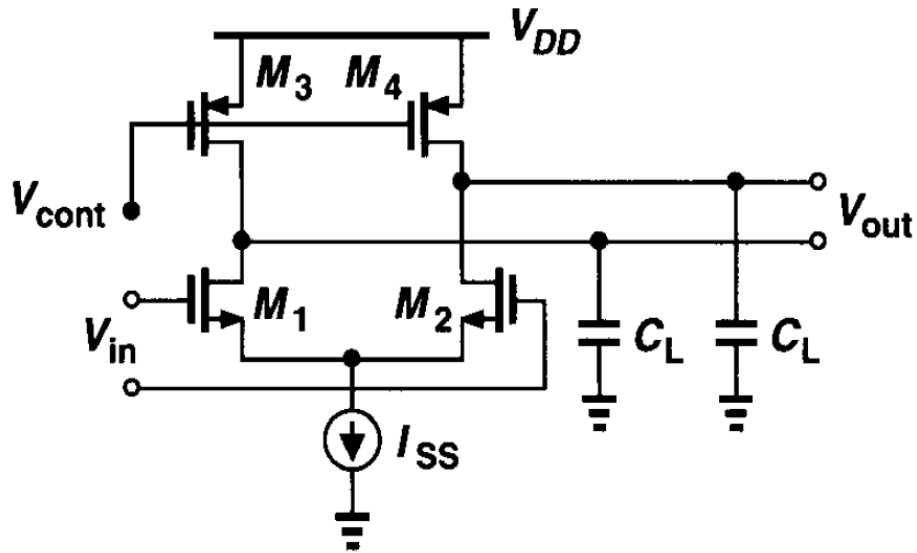
- $\tan^{-1} \frac{\omega_{osc}}{\omega_{RC}} = \frac{180^\circ}{N}, \omega_{osc} = \omega_{RC} \tan\left(\frac{180^\circ}{N}\right)$

- $|H(s = j\omega_{osc})| = 1 \rightarrow \frac{A_0^N}{\left(\sqrt{1 + \left(\frac{\omega_{osc}}{\omega_{RC}} \right)^2} \right)^N} = 1$

- (1) and (2) $\rightarrow A_0 = \sqrt{1 + \tan^2\left(\frac{180^\circ}{N}\right)}$



Ring Oscillator Delay Cells and Frequency Tuning



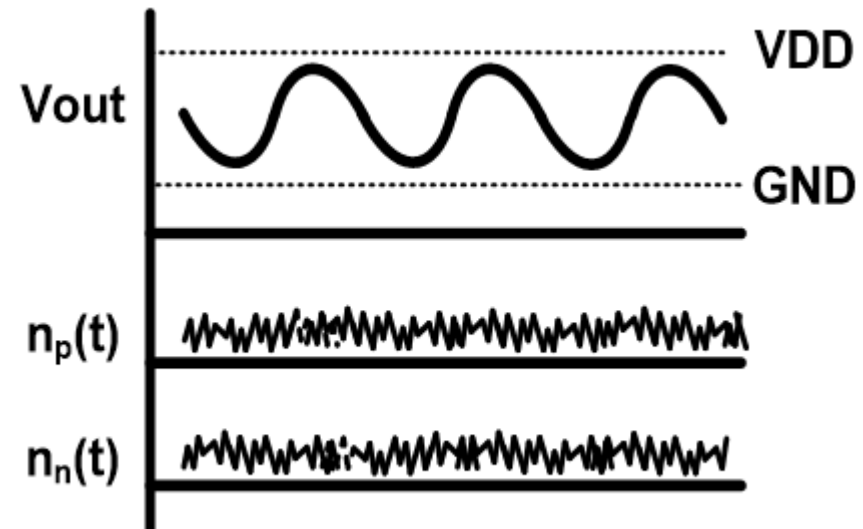
- Unsaturated delay cell
- Output swing is close to sinusoidal
- All transistors are on and contribute noise
- Tail current noise

- Ring oscillator oscillation frequency is determined by the RC time constant of each delay cell

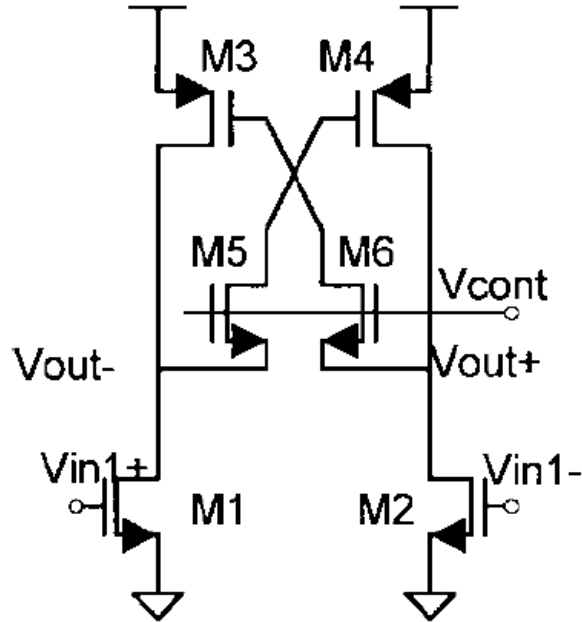
$$\tau_1 = R_{on3,4} C_L$$

$$= \frac{C_L}{\mu_p C_{ox} \left(\frac{W}{L}\right)_{3,4} (V_{DD} - V_{cont} - |V_{THP}|)}$$

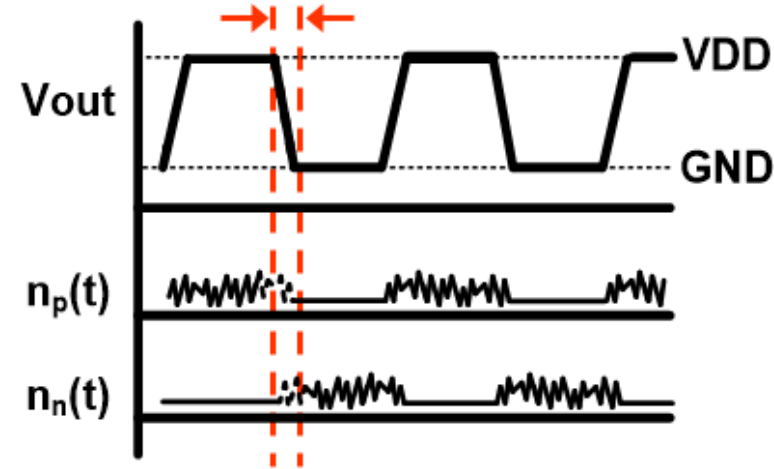
- Control R_{on} through the gate voltage of $M_{3/4}$
- C_L can be implemented as varactor whose capacitance depends on its tuning voltage.



Ring Oscillator Delay Cells and Frequency Tuning



- M3 and M4 form a latch
- M5 and M6 controls the strength of the latch
 - Higher Vcont, stronger latch, more delay, lower oscillation frequency



- Saturated delay cell
- Output swing is rail to rail, close to square wave.
- Noise only from transistors during turned-on time
- No tail current flicker noise upconversion

Ref [11])



Ring Oscillator and LC Oscillator Comparison

	Ring oscillator	LC oscillator
Phase noise	High	Low due to high Q LC tank
Area	Small	Large due to on-chip inductor
Tuning range	Wide by tuning both driving current and loading capacitance	Medium
Figure of Merit	Low: high phase noise at same current consumption	High: low phase noise at same current consumption
Applications	Clock generation, clock data recovery, multiphase generation	Wireless transceiver frequency synthesis

- Ring oscillators are widely used in clock generation and CDR (clock and data recovery). Compact size but poor noise performance.
- LC oscillators are used in frequency synthesizers where phase noise performance is critical.



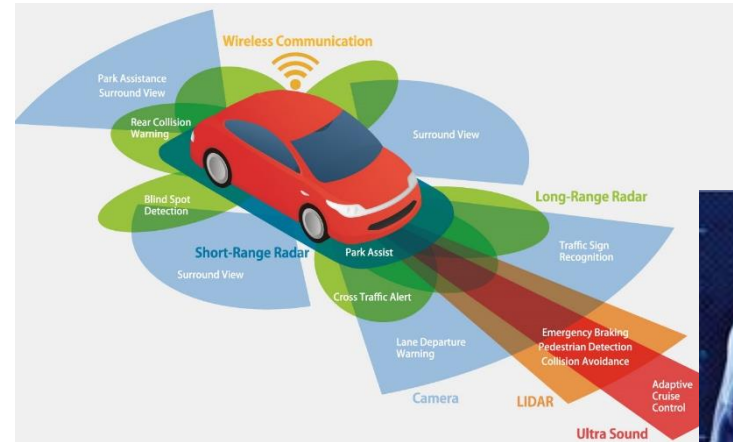
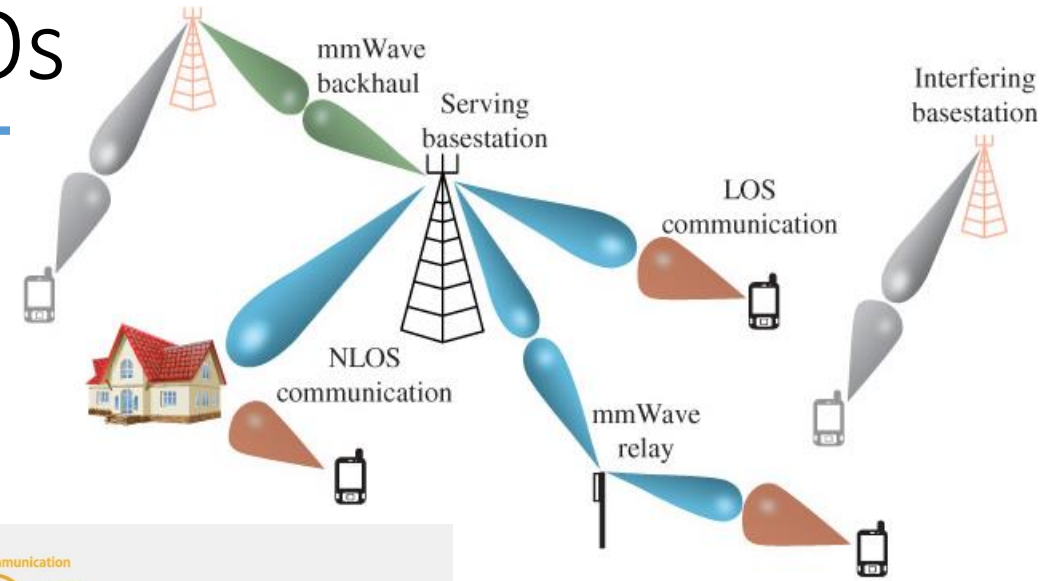
Couple Oscillator Summary

- In phase coupling
 - Phase noise performance enhancement
 - Dual core or multi-core
- Anti-phase coupling
 - Quadrature oscillator
- Multi-phase coupling
 - Multi-phase oscillator
 - LC based or inductor-less ring oscillator
- Ring oscillator applications
 - Frequency generation
 - Multi-phase generation
 - Time measurement (e.g. time to digital converter)



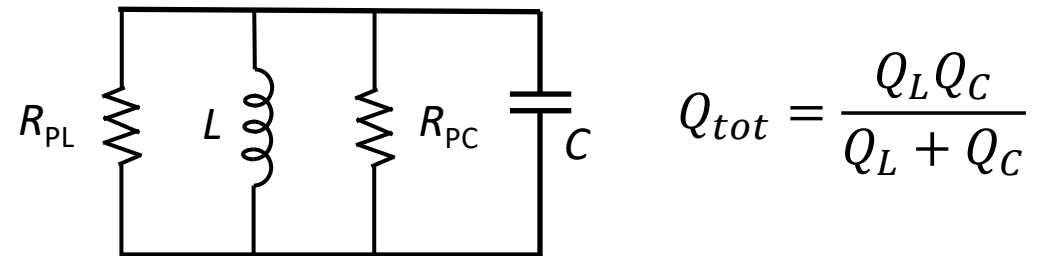
Millimeter Wave (mmW) VCOs

- Common millimeter wave frequency bands
 - mmW communication applications
 - 5G FR2 (frequency range 2)
 - n257: 26.5 – 29.5GHz
 - n258: 24.25 – 27.5GHz
 - n258: 39.5 – 43.5GHz
 - n260: 37.0 – 40.0GHz
 - n261: 27.5 – 28.35GHz
 - FYI 5G FR1: 600MHz – 7.1GHz
 - Satellite communication: 130MHz – 35GHz
 - 802.11ad WLAN 57 – 64GHz
 - Automotive radar applications
 - ISM band and ultra wide band (UWB): 21 – 26GHz
 - Long range radar and short range radar bands: 76 – 81GHz
 - Image sensor: 50 – 700GHz
 - Terahertz spectroscopy: 300GHz – 30THz

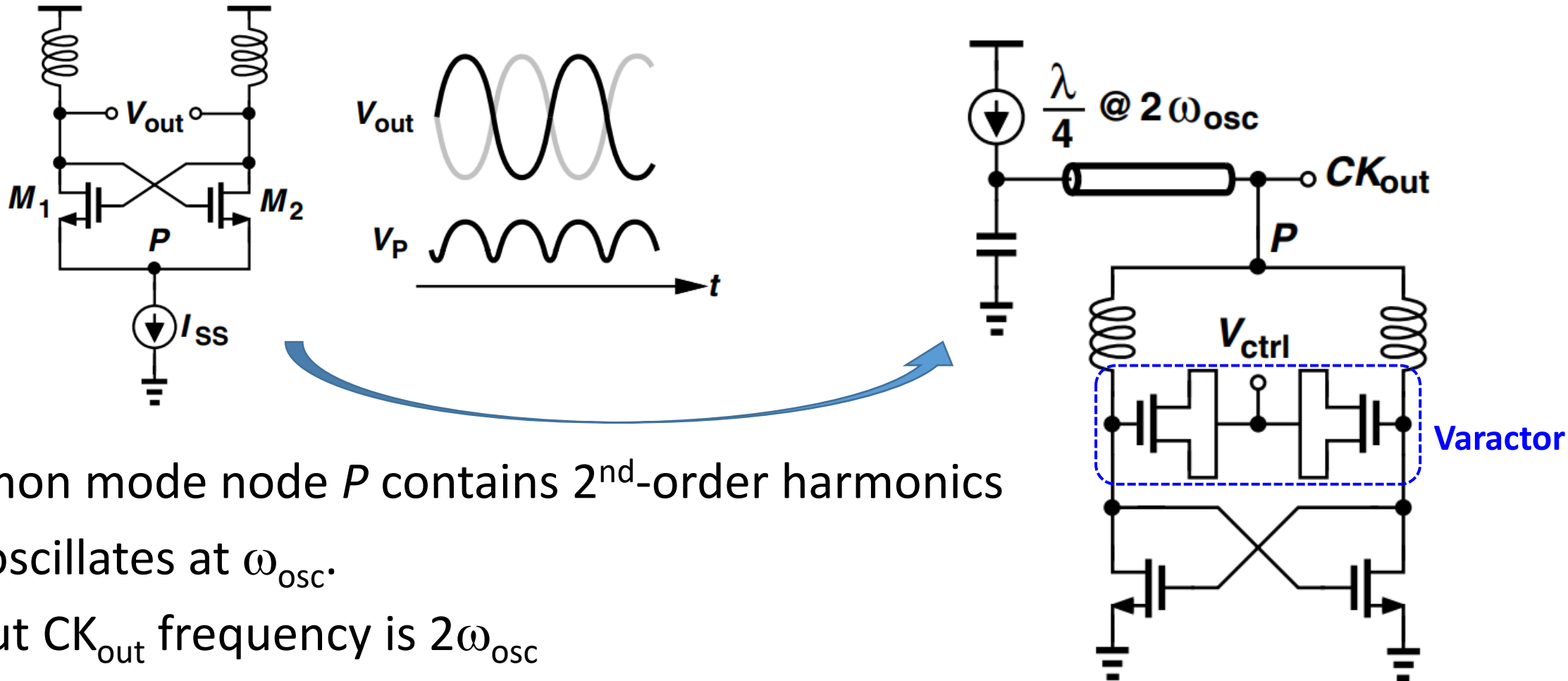


Challenges and Techniques in mmW VCO Design

- Challenges
 - Oscillation frequency becomes closer to transistor f_T and $f_{\max} \rightarrow$ startup condition
 - Parasitic capacitance and inductance becomes a larger percentage of tank capacitance and inductance $\rightarrow$ frequency tuning range
 - Tank Q dominated by tank capacitors, both switched capacitor bank and varactor $\rightarrow$ startup condition and phase noise performance
- Design techniques
 - Frequency multiplier to allow VCO oscillate at lower frequency
 - Harmonic oscillator (push-push, multi-push VCOs)
 - Different tuning schemes, such as interpolative tuning, magnetic tuning
 - Distributed oscillator
 - EM modeling technique and accuracy



Push-Push Cross-Coupled g_m VCO

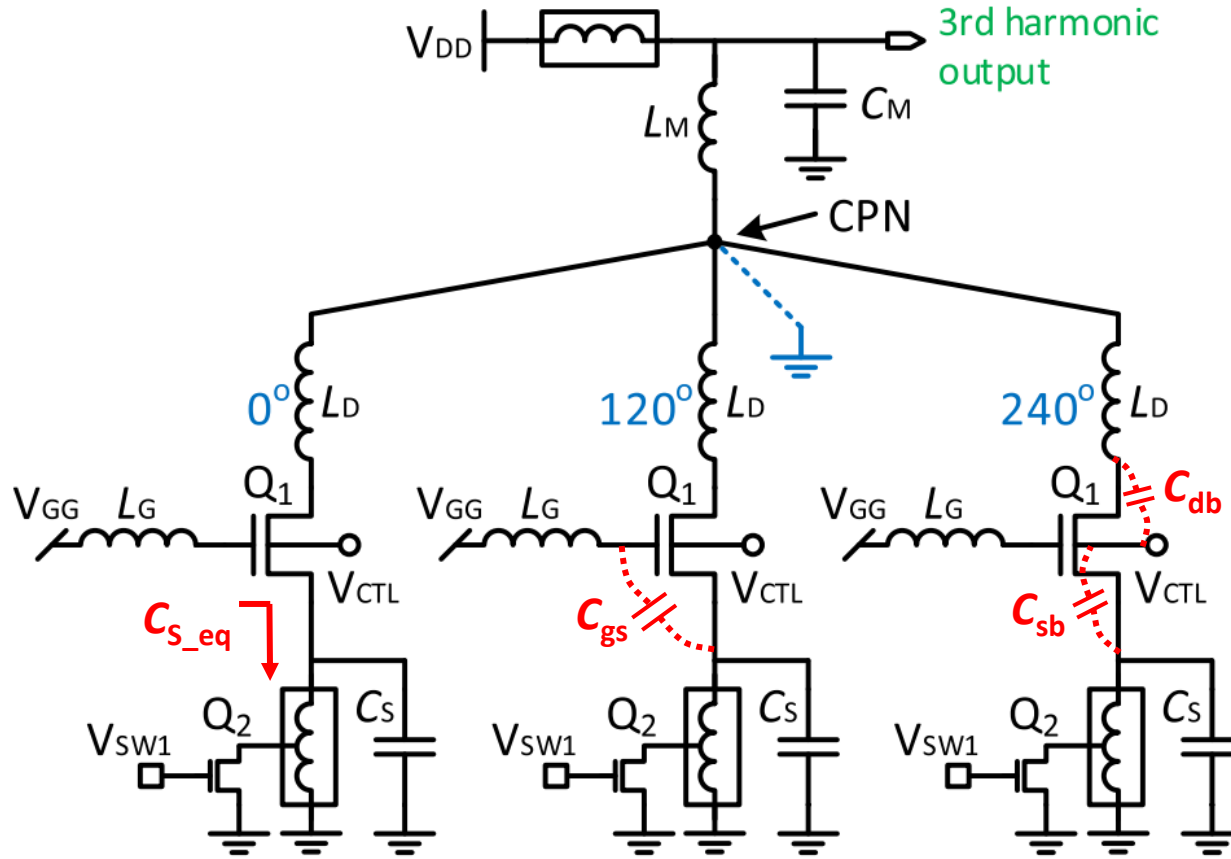


- Common mode node P contains 2nd-order harmonics
- VCO oscillates at ω_{osc} .
- Output CK_{out} frequency is $2\omega_{osc}$
- $\lambda/4$ transmission line short at DC and open at $2\omega_{osc}$

[Ref 12]



Triple-Push Colpitts VCO



- 560GHz VCO, 20GHz tuning range
- 65nm CMOS ($f_{\max} = 240\text{GHz}$)
- Each Colpitts VCO oscillates at ω_{osc} .
- Three VCOs are 120° phase shifted from each other
- Common push node CPN terminated with a 3rd-harmonic tank $L_M // C_M$
- Output frequency is $3\omega_{\text{osc}}$
- Coarse tuning by V_{SW1} : change C_{S_eq}
- Fine tuning by V_{CTL} : change $C_{sb}, C_{db} \rightarrow C_{S_eq}$

- Colpitts tank: L_G, C_{gs} and C_{S_eq}
- Colpitts positive feedback: C_{gs} and C_S

$$\omega_{\text{osc}} = \frac{1}{\sqrt{L_G \frac{C_{gs} C_{S_eq}}{C_{gs} + C_{S_eq}}}}$$

[Ref 13]



References

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- 2) S. Amir-Reza, et al., Analysis and design of a multi-core oscillator for ultra-low phase noise, TCAS-I, 2016
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